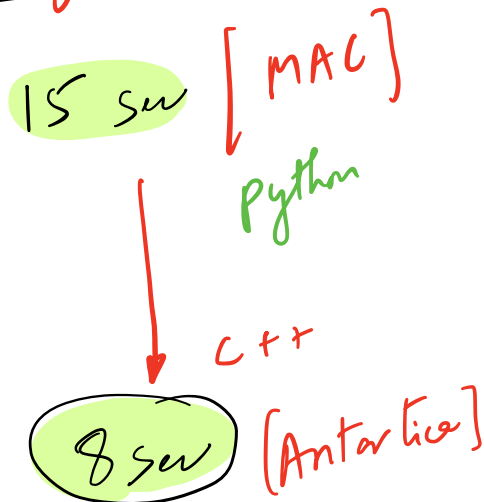


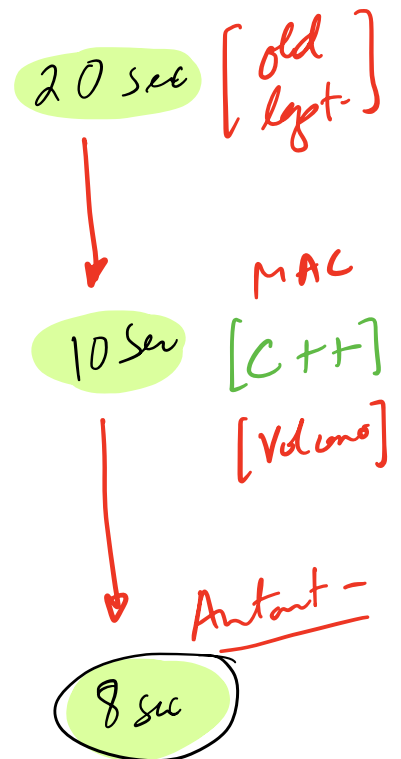
Time Complexity - 2

1) Given 10^4 no's., write an algo to sort these nos.

Surge 



Sumanth 



④ Execution time is not a good factor

↳ S/w + H/w + outside factors

$s=0, p=0;$

$\{ (i=0; i < N; i++) \{$

$s++;$ ✓ $\rightarrow$

$p--;$ ✓

$\}$

$\rightarrow$ #it $\rightarrow N$

#ops $\rightarrow \sim 4N$

Con

Given N elements to sort!

Athul

$100 \lg_2 N$

V_s

Ajay

$N/10$

ops

$N = 4$

$100 \cdot \lg_2 4$

$\downarrow$
 $100 \times 2 = \underline{200 \text{ ops}}$

$4/10$

$\approx 0.4 \text{ ops}$

$N = 10^3$

$100 \cdot \lg_2 10^3$

$100 \cdot \lg_2 2^{10}$

1000 ops

$1000/10$

100 ops

$$N = 10^6$$

$$100 \cdot \log_2 10^6$$

$$100 \log_2 2^{20}$$

$$2000 \text{ ops}$$

$$\frac{10^6}{10}$$

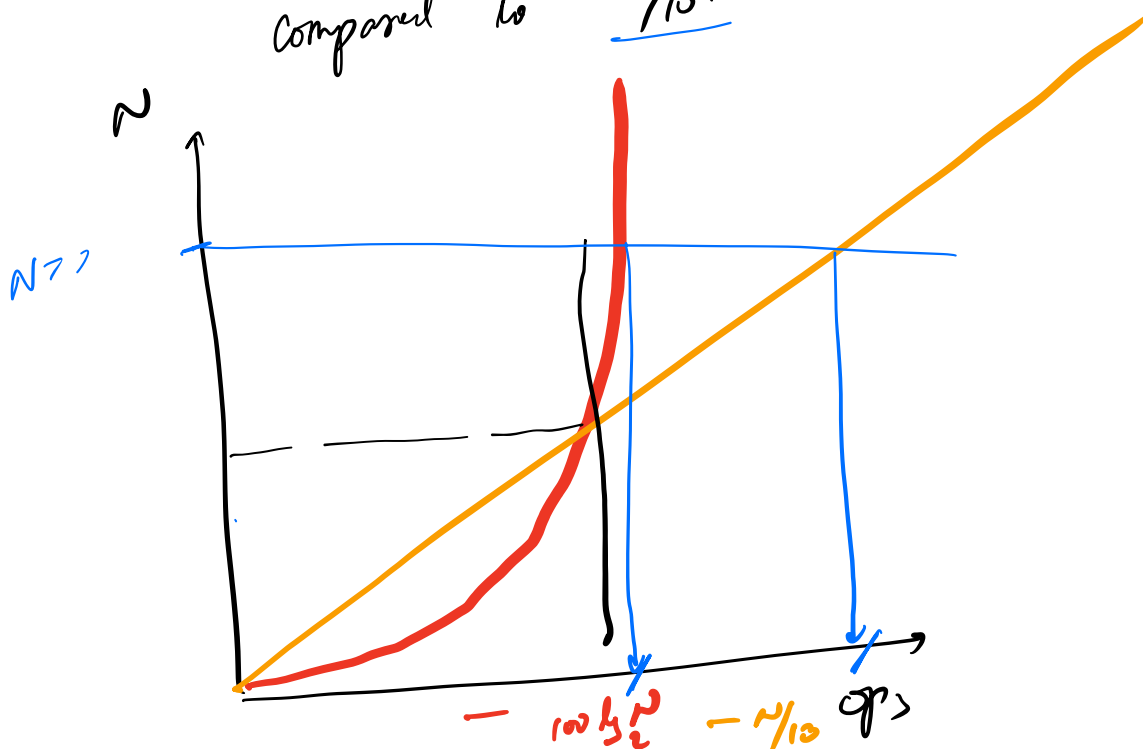
$$= 10^5 \text{ ops}$$

$$N = 10^9$$

$$= 3000 \text{ ops}$$

$$10^8 \text{ ops}$$

* $100 \log_2 N$ would perform worse for smaller values of N compared to $N/10$.



⑥ Algo 1 , Algo 2 , Algo 3 . . . Algo 5.

Input size is large for real applications!

<u>Hotstar</u>	<u>YT</u>	<u>fb</u>	<u>inst</u>
2×10^7	1×10^3	3×10^3	2×10^9

A Symptotic Analysis of Algo's

→ judge the performance of your algo for very large input values!

Single ops +, -, > . .

*, / , %

→ 10^8 ops in 1 sec.

$100 \log_2 N$

N^3

$$\underline{N = 10^6}$$

$$\downarrow 100 \cdot \frac{1}{2} 10^6$$

$$\underline{2000 \text{ ops}}$$

$$\sim \frac{2000 \text{ sec}}{10^8}$$

$$\downarrow \underline{10^{18} \text{ ops}}$$

$$\frac{10^{18}}{10^8} \text{ sec}$$

$$\underline{10^{10} \text{ sec}}$$

$$\downarrow \underline{317 \text{ yrs}}$$

① Big O Notation → for judging algs.

- ✓ 1) Calculate the # ops. —
- ✓ 2) Neglect lower order terms!
- ✓ 3) Neglect the constant coeff.

$$\frac{1}{2}N < \sqrt{N} < N < N \frac{1}{2}N < N\sqrt{N} < N^2 < N^3 \text{ — } < 2^N$$

$$Q \quad f(N) = 4N + \underbrace{3N \log N}_{3N \log N} + 1$$

$$3N \log N$$

$$TC = O(N \log N)$$

$$Q \quad f(N) = 4N \log N + \underbrace{3N \sqrt{N}}_{3N \sqrt{N}} + 10^6$$

$$N \rightarrow 10^6$$

$$TC = O(N \sqrt{N})$$

Why to neglect lower order terms →

$N^2 + 10N$	$4N^2$	$10N$
$N = 10$	100 50%	100 50%
$N = 100$	$\frac{10^4}{10^4 + 10^3} \times 100\%$ $\sim 90\%$	$\frac{10^3}{10^4 + 10^3} \times 100\%$ $\sim 10\%$
$N = 10^6$	$\frac{10^{12}}{10^{12} + 10^7} \times 100\%$ $\sim 99.9\%$	$\frac{10^7}{10^{12} + 10^7} \times 100\%$ $\sim 0.001\%$

① Sort N no's $\rightarrow$

A

B

q_{u1}

$$10 \log_2 N$$

$$O(\log N)$$

$$N \quad O(N)$$

q_{u2}

$$10^4 \log_2 N$$

$$O(\log N)$$

$$N \quad O(N)$$

3

$$10^6 N \log_2 N + N^2$$

$$O(N^2)$$

$$2N^2$$

$$O(N^2)$$

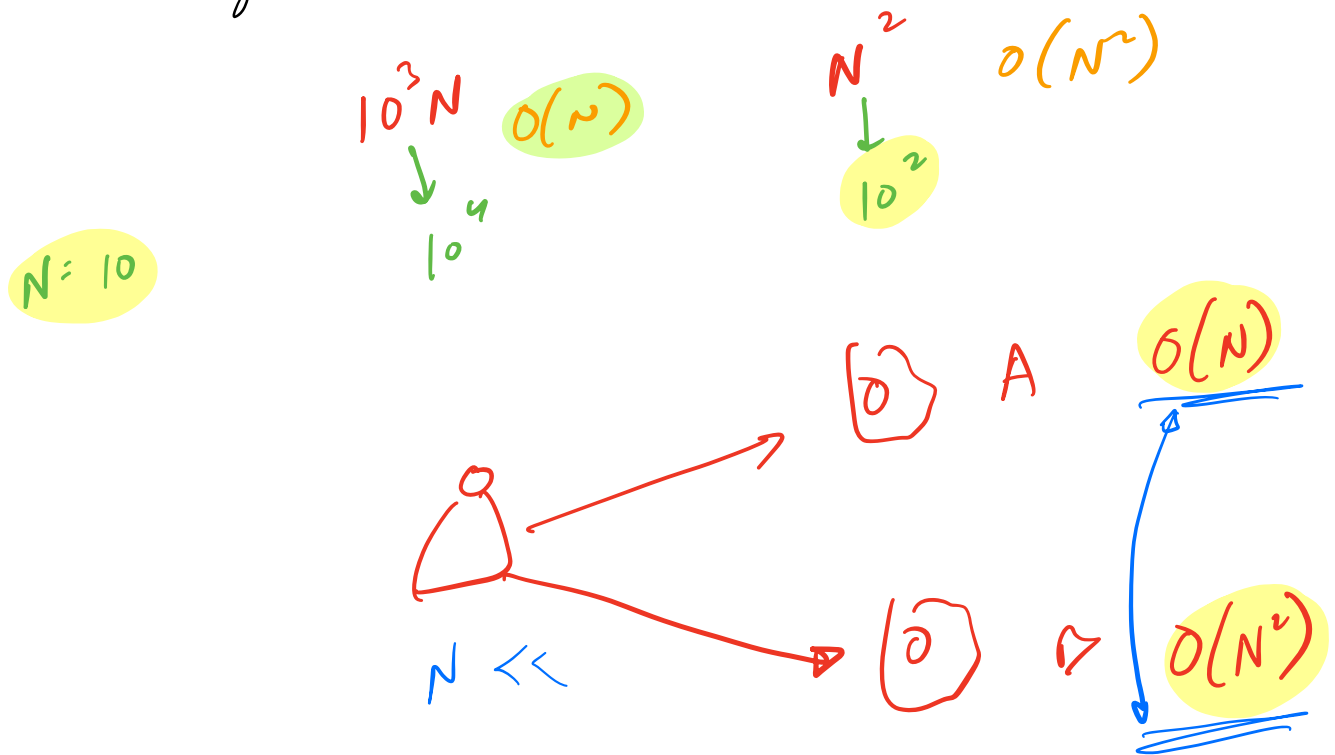
4

$$10N^2 + 5N$$

$$5N^2 + 10$$

Issues with Big O

1) For smaller input values Big O doesn't give us a fair idea:



2) If 2 Algo have same highest order term

$$N^2 \quad O(N^2) \quad \underline{\underline{10^6 N^2}} \quad O(N^2)$$

Space Complexity

Big O

```
fun (int N) {  
  int x = N; → 4B  
  int y = 10; → 4B  
  long z = 100; → 8B  
  double w = 10.0; → 8B  
}
```

int	→	4B
long	→	8B
double	→	8B

$$f(N) = 24 N^0$$

SC = O(1)
↓
Constant!

```

fun (int n) {
    int x = n; → 4B
    int y = n; → 4B
    long z = 10; → 8B
    int b[10]; → 40B
    int arr[N]; → 4NB
}

```

$$f(N) = 4N + 56$$

$$SC: O(N)$$

```

fun (int n) {
    int x = n; → 4B
    int y = n; → 4B
    long z = 10; → 8B
    int arr[n]; → 4NB
    int b[n][n]; → 4N²B
}

```

$f(N) = 20 + 4N^2$

$$SC = O(N^2)$$

Q Given N no's - find their sum $\rightarrow$

```
int findSum ( int arr[ ], int n ) {  
    int sum = 0;
```

```
    for ( i: 0  $\rightarrow$  n ) {  
        sum = sum + arr[i];
```

```
    }  
    return sum;  
}
```

$TC = O(N)$

$SC = O(1)$

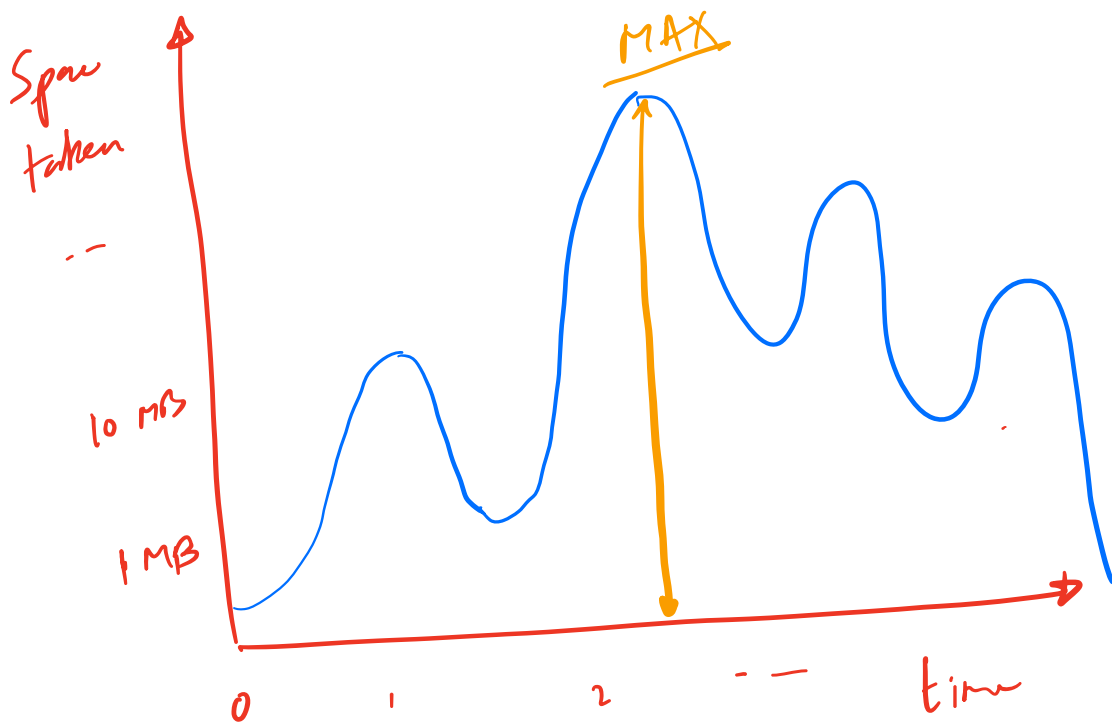
neglect this

* Space Complexity $\rightarrow$ Amount of Extra Space taken by your algo other than the input & output

```
for ( i: 0  $\rightarrow$  n ) {  
    int n = 10;  
}
```

$SC = O(1)$

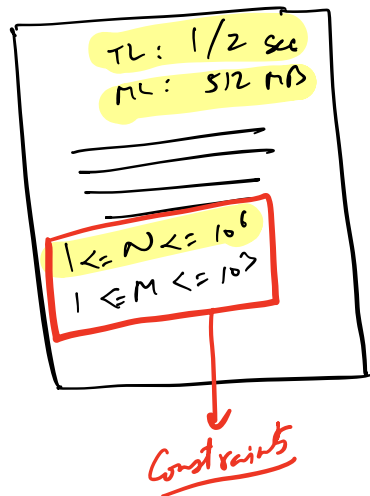
① SC is the MAX amt of space taken by algo at any point in time!



TLE : Time Limit Exceeded !

1) $N^2 \rightarrow (10^6)^2 \rightarrow 10^{12} \text{ ops}$
 $\sim 10^4 \text{ sec}$ TLE

2) $N\sqrt{N} \rightarrow 10^6 \sqrt{10^6} = 10^9 \text{ ops}$
 $\sim 10 \text{ sec}$ TLE



3) $100 N \rightarrow 100 \cdot 10^6 = 10^8 \text{ ops}$
 $\sim 1 \text{ sec}$ ✓

4) $N \log N \rightarrow 10^6 \cdot \log 10^6 \quad 10^6 \cdot 20$
 $2 \times 10^7 \text{ ops}$
 $< 1 \text{ sec}$ ✓✓✓

4) $N \rightarrow 10^6 \text{ ops}$
 $< 1 \text{ sec}$ ✓✓✓✓

Take Constraints very Seriously

$N \leq 100$ / (N^3)

$1 \leq N \leq 20 \rightarrow 2^N$

$1 \leq N \leq 1000$

$\rightarrow N^2, N^2 \log N, N, N \log N, N \sqrt{N}$
 (Note: $N \sqrt{N}$ has 'x' marks below it)

$1 \leq N \leq 10^5, 10^6$

$1 \leq N \leq 10^7$

$\rightarrow N, N \log N$

$1 \leq N \leq 10^9$

$\rightarrow \sqrt{N}, \log N$

$1 \leq N \leq 10^{18}$

$\rightarrow \log N$